A MINI PROJECT REPORT ON

COVID-19 Vaccination Data Analysis

SUBMITTED TO

SAVITRIBAI PHULE PUNE UNIVERSITY, PUNE

SUBMITTED BY

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SINHGAD TECHNICAL EDUCATION SOCIETY-PUNE

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Sinhgad Institutes

UNDER THE GUIDANCE OF

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CERTIFICATE

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This is to certify that the project report entitled **COVID-19 Vaccination Data Analysis** submitted by

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is a bonafide work carried out under our supervision and guidance. This work is submitted in partial fulfillment of the requirements of Data Science and Big Data Analysis Laboratory of Savitribai Phule Pune University, Pune.

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Date: ____________

Place: Pune

ACKNOWLEDGEMENT

It gives me great pleasure to present this project report on **COVID-19 Vaccination Data Analysis Using Python**. I express sincere gratitude and thanks to my guide **Prof. M. P. Navale** for constant guidance, motivation, and valuable suggestions throughout the mini project work.

I am thankful to the Head of Department, all faculty members, and the institute for providing the required infrastructure and support. I also thank my family and friends for their encouragement and support during the completion of this project.

ABSTRACT

This mini project presents a **COVID-19 vaccination data analysis** system implemented in Python using the Pandas and Matplotlib libraries. The system uses a state-wise vaccination dataset containing information about first dose, second dose, and gender-wise vaccination across India.

The project follows a data analysis approach where important attributes such as state, first dose, second dose, and gender-based vaccination (male and female) are processed and analyzed. Data preprocessing techniques such as handling missing values, cleaning column names, and transforming data into structured format are applied to ensure accurate results.

State-wise aggregation is performed to compute the total number of people vaccinated with the first and second doses. Additionally, gender-wise analysis is carried out to determine the total number of males and females vaccinated. Visualization techniques such as bar charts and pie charts are used to represent the data effectively.

The entire workflow is implemented in a Jupyter Notebook for easy execution, visualization, and understanding. The project demonstrates practical use of data analysis concepts, data preprocessing, aggregation, and visualization techniques on a real-world dataset.

Contents

Chapter 1

Introduction

1.1 Project Title

COVID-19 Vaccination Data Analysis Using Python

1.2 Introduction

Data analysis plays a crucial role in understanding large datasets and extracting meaningful insights. During the COVID-19 pandemic, vaccination data became essential to monitor the progress of immunization programs across different regions.

This project focuses on analyzing COVID-19 vaccination data in India using Python. The dataset contains state-wise vaccination details including first dose, second dose, and gender-wise vaccination statistics. By applying data analysis techniques, meaningful insights such as total vaccinations and comparisons across states can be obtained.

The implementation is carried out in a Jupyter Notebook using Python libraries such as Pandas and Matplotlib, making it easy to analyze, visualize, and interpret the data.

1.3 Objectives of the Project

- To analyze COVID-19 vaccination data using Python
- To calculate state-wise first dose vaccination
- To calculate state-wise second dose vaccination
- To determine total male and female vaccinated
- To visualize vaccination data using graphs.

1.4 Scope

The project focuses on analyzing vaccination data and generating insights. It can be extended further by:

- Adding real-time data updates
- Creating dashboards using Power BI or Tableau
- Performing predictive analysis using machine learning

Chapter 2

Literature Review

Data analysis has become an essential tool in modern decision-making. Various tools and technologies such as Python, Pandas, and data visualization libraries are widely used to process and analyze large datasets.

In the healthcare domain, data analysis helps in understanding disease patterns, vaccination trends, and resource allocation. COVID-19 vaccination datasets have been widely analyzed to track progress and improve decision-making.

This project uses basic data analysis techniques such as grouping, aggregation, and visualization to extract meaningful insights from vaccination data.

Chapter 3

Software Requirement Specification

3.1 Functional Requirements

- Load COVID-19 vaccination dataset from covid_vaccine_statewise.csv.
- Clean and preprocess data (handle missing values, remove extra spaces in column names).
- Extract relevant columns such as State, First Dose, Second Dose, Male, and Female.
- Perform state-wise aggregation using groupby operations.
- Calculate total number of people vaccinated with first and second doses.
- Compute total number of male and female vaccinated individuals.

3.2 Non-Functional Requirements

• Usability: The system should have a simple and easy-to-understand notebook workflow with clear outputs.
• Performance: The system should process the dataset efficiently and provide quick results.
• Maintainability: The code should be modular and easy to modify for future enhancements.
• Reliability: The system should handle missing values and incorrect inputs gracefully.
• Reproducibility: The system should produce consistent results for the same dataset and inputs.
• Scalability: The system should be able to handle larger datasets with minimal changes.

3.3 Hardware Requirements

- Processor: Intel Core i3 or above
- RAM: 4 GB or above (8 GB recommended)

- Storage: 1 GB free disk space

3.4 Software Requirements

- Operating System: Windows 10/11 or Linux
- Python: 3.10 or above
- Notebook Environment: Jupyter Notebook / VS Code Notebook
- Libraries: pandas, scikit-learn
- Optional IDE: VS Code

Chapter 4

System Analysis and Design

4.1 Existing System

Traditional methods of analyzing vaccination data involve manual calculations or the use of basic spreadsheet tools. These methods are time-consuming, prone to errors, and inefficient when dealing with large datasets. Extracting insights such as state-wise vaccination or gender-wise distribution requires significant effort and lacks automation.

4.2 Proposed System

The proposed system analyzes COVID-19 vaccination data using Python and data analysis libraries. It automates the process of data cleaning, aggregation, and visualization.

Key advantages are:

- Automated data processing and analysis
- Accurate and fast computation of vaccination statistics
- Easy visualization using graphs
- Simple implementation using Python and Jupyter Notebook
- Scalable for larger datasets and future enhancements

4.3 Dataset Overview

Dataset file used:

- covid_vaccine_statewise.csv

Important fields considered:

- State
- First Dose
- Second Dose
- Male (Individuals Vaccinated)
- Female (Individuals Vaccinated)

Additional fields available:

- Total Doses Administered
- Age group vaccination data
- Vaccine type data (Covaxin, Covishield, etc.)

4.4 Recommendation Model

The dataset is processed using data analysis techniques. Relevant columns are selected and cleaned to remove inconsistencies. State-wise aggregation is performed using grouping functions to calculate total vaccinations.

Basic computation used:
Total Vaccination = First Dose + Second Dose

State-wise aggregation:

Total First Dose (State) = Σ First Dose
Total Second Dose (State) = Σ Second Dose

Gender-wise totals:
Total Male = Σ Male (Individuals Vaccinated)
Total Female = Σ Female (Individuals Vaccinated)

Visualization techniques such as bar charts and pie charts are used to represent the data effectively.
The system provides clear insights into vaccination distribution across different states and genders

Chapter 5

System Design and Implementation

5.1 Implementation Workflow

The notebook implementation follows this sequence:

- Import required Python libraries such as Pandas and Matplotlib
- Load the vaccination dataset from a local CSV file
- Clean and preprocess the dataset (handle missing values, fix column names)
- Select relevant columns for analysis
- Perform state-wise aggregation using groupby operations
- Calculate total first dose and second dose vaccinations
- Compute total male and female vaccinated individuals
- Generate visualizations such as bar charts and pie charts
- Display results in the Jupyter Notebook

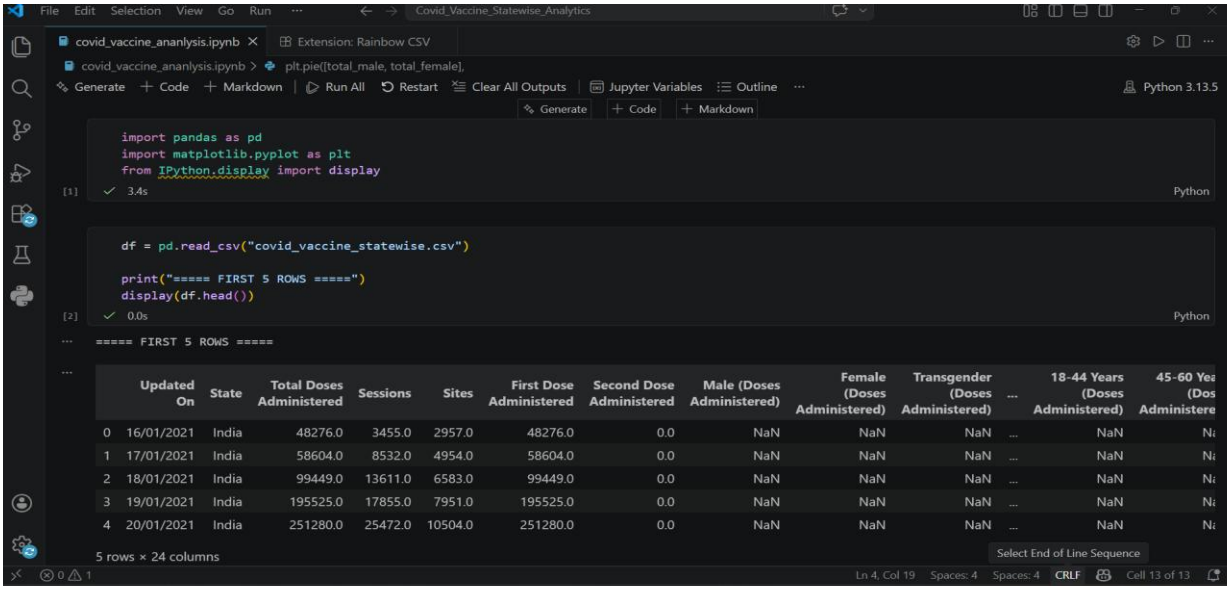
5.2 Notebook and Project Files

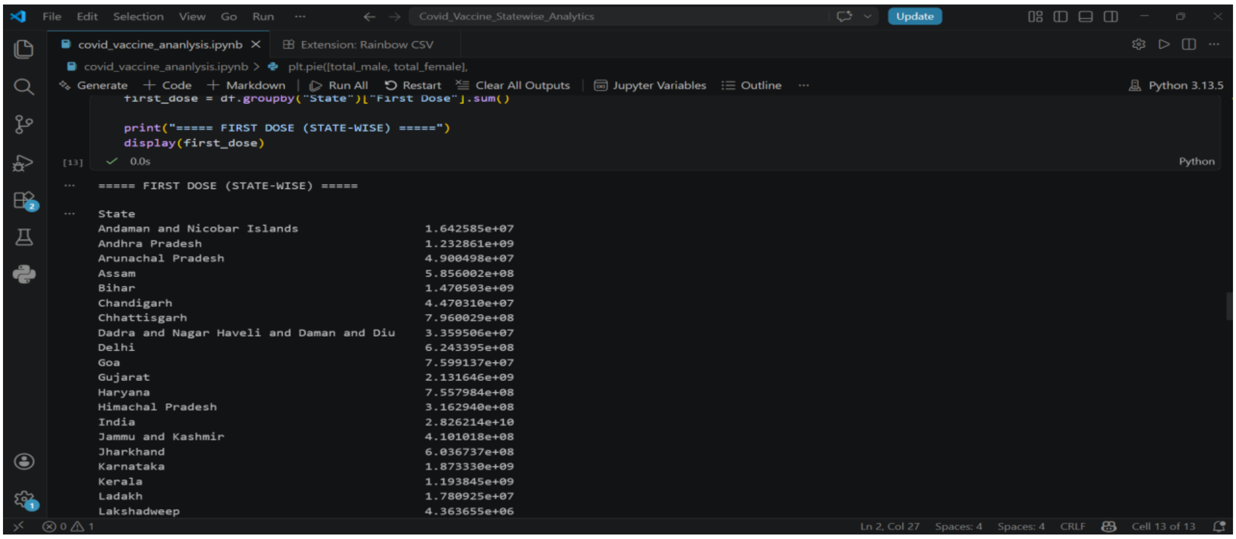
- vaccination_analysis.ipynb – Main implementation
- covid_vaccine_statewise.csv – Dataset
- requirements.txt – Python dependencies (pandas, matplotlib)

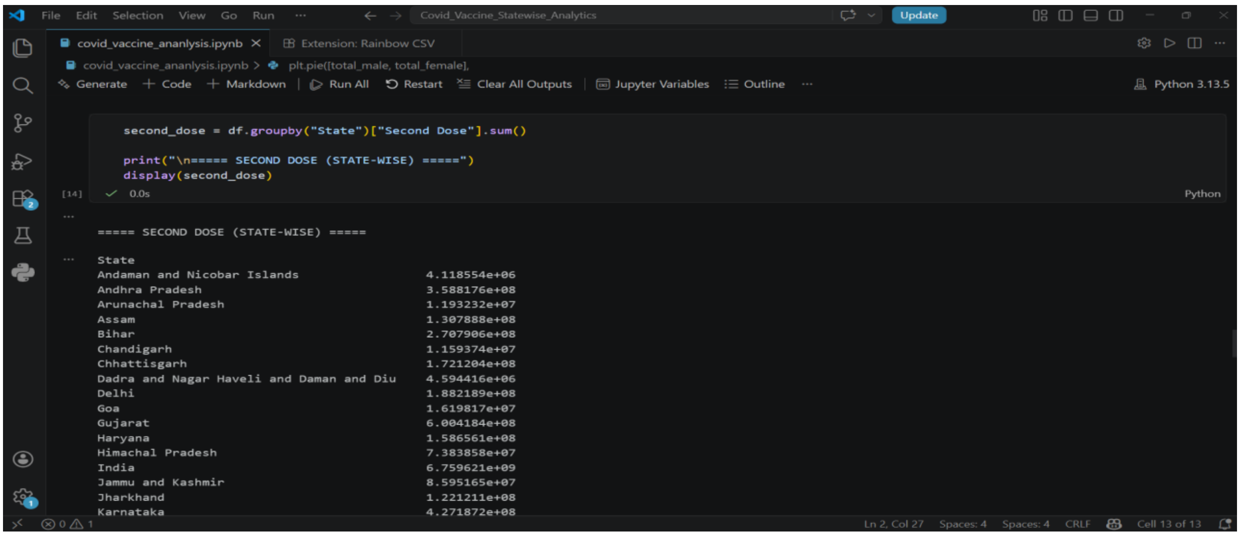
5.3 Algorithmic Highlights

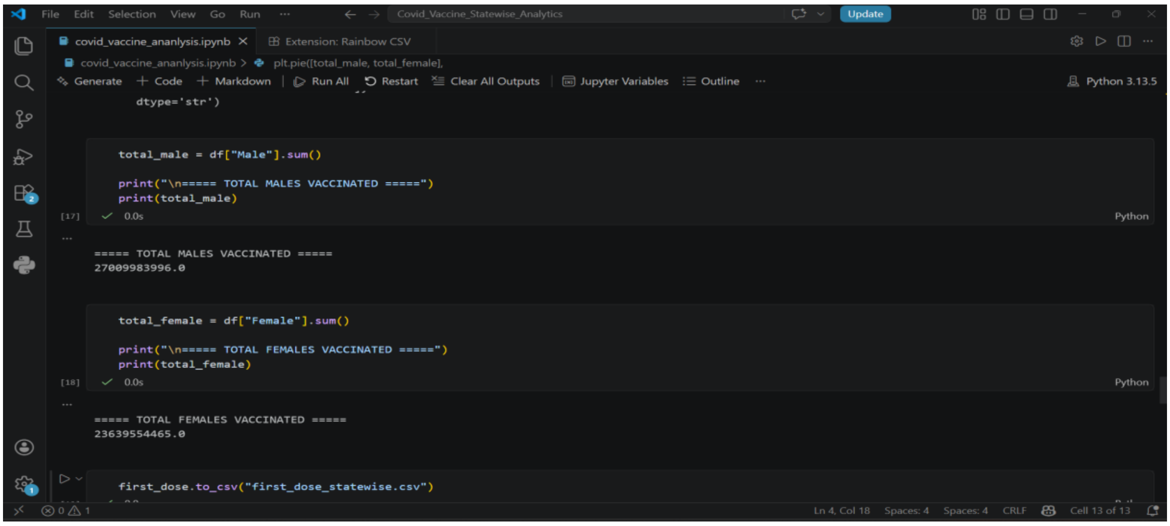
- Used Pandas library for efficient data manipulation and analysis
- Applied groupby operations for state-wise aggregation
- Used sum() function to calculate total vaccination values
- Performed data cleaning to handle missing values and inconsistent column names
- Implemented visualization techniques for better understanding of data
- Generated ranked outputs such as top states based on vaccination

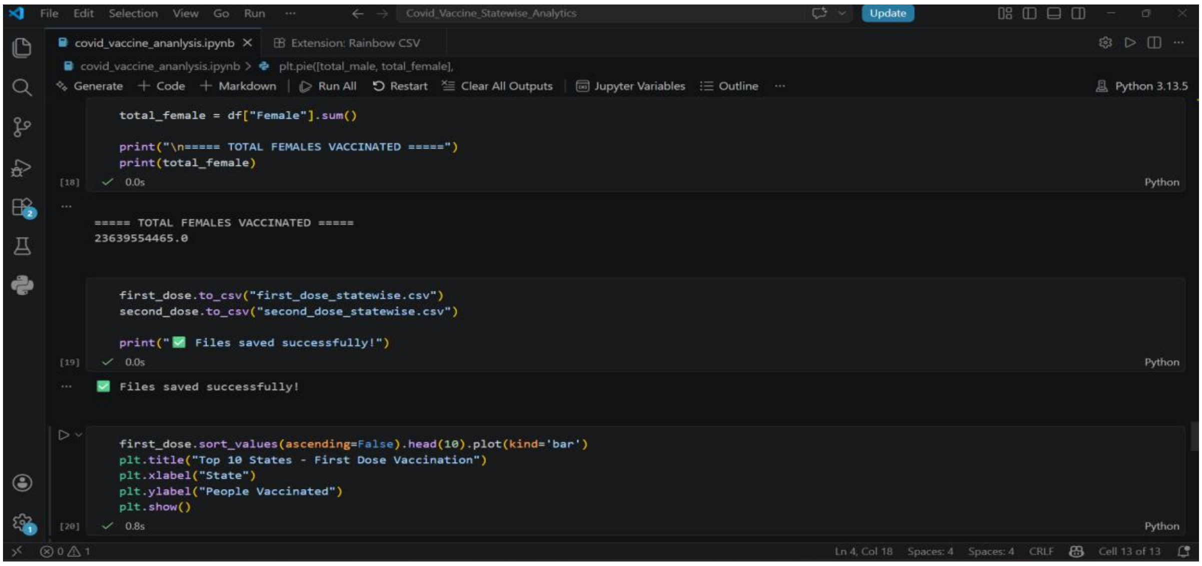
5.4 Sample Output

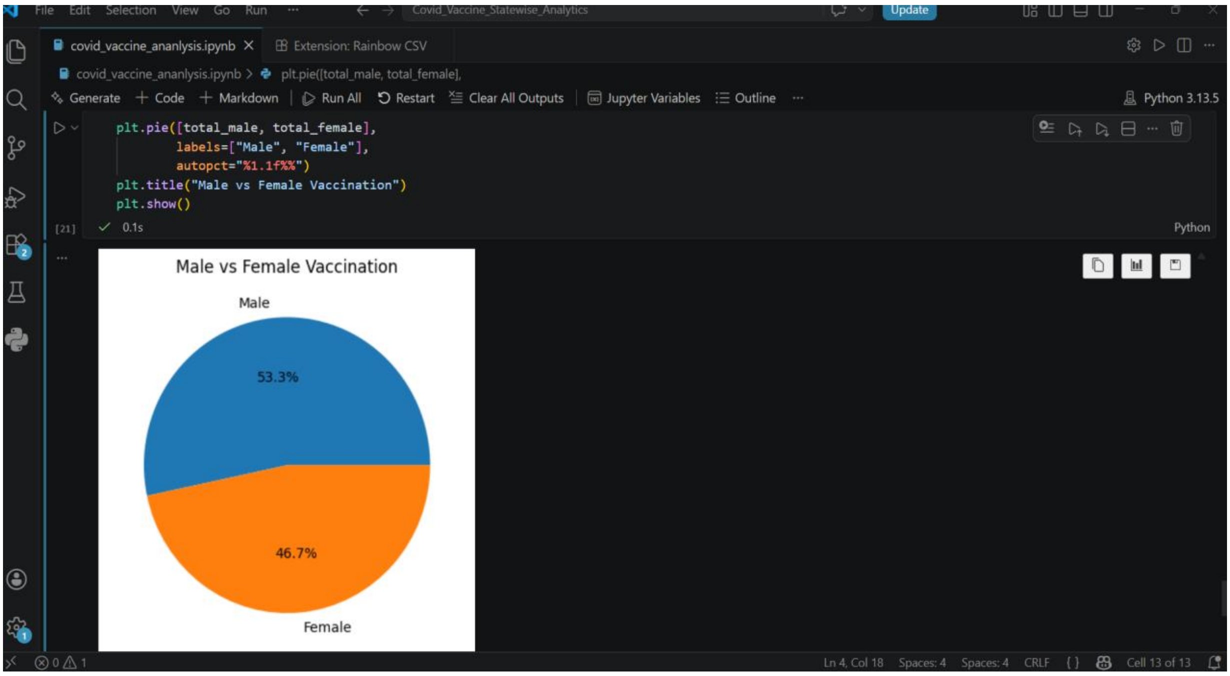







covid_vaccine_ananlysis.ipynb
Extension: Rainbow CSV
dtype='str')
total_male = df["Male"].sum()
print("\n===== TOTAL MALES VACCINATED =====")
print(total_male)
===== TOTAL MALES VACCINATED =====
27009983996.0
total_female = df["Female"].sum()
print("\n===== TOTAL FEMALES VACCINATED =====")
print(total_female)
===== TOTAL FEMALES VACCINATED =====
23639554465.0
first_dose.to_csv("first_dose_statewise.csv")


covid_vaccine_ananlysis.ipynb
Extension: Rainbow CSV
total_female = df["Female"].sum()
print("\n===== TOTAL FEMALES VACCINATED =====")
print(total_female)
===== TOTAL FEMALES VACCINATED =====
23639554465.0
first_dose.to_csv("first_dose_statewise.csv")
second_dose.to_csv("second_dose_statewise.csv")
print("✅ Files saved successfully!")
✅ Files saved successfully!
first_dose.sort_values(ascending=False).head(10).plot(kind='bar')
plt.title("Top 10 States - First Dose Vaccination")
plt.xlabel("State")
plt.ylabel("People Vaccinated")
plt.show()


covid_vaccine_ananlysis.ipynb
Extension: Rainbow CSV
plt.pie([total_male, total_female],
labels=["Male", "Female"],
autopct="%1.1f%%")
plt.title("Male vs Female Vaccination")
plt.show()
Male vs Female Vaccination
Male
53.3%
46.7%
Female

Chapter 6

Testing

6.1 Testing Techniques Used

- **Functional Testing:** Validation of each step in the notebook such as data loading, cleaning, aggregation, and visualization
- **Data Validation Testing:** Checking dataset integrity, file existence, and correctness of column names before processing
- **Result Verification:** Ensuring correctness of state-wise vaccination totals and gender-wise calculations
- **Error Handling Testing:** Handling scenarios such as missing columns, incorrect file paths, and null values in the dataset
- **Visualization Testing:** Verifying that graphs such as bar charts and pie charts are generated correctly and display accurate information

6.2 Sample Test Cases

TC ID	Test Case Description	Input	Expected Output	Status
TC01	Load dataset	covid_vaccine_statewise.csv file	Dataset loaded successfully without errors	Pass
TC02	Check column names	Dataset columns	Required columns (State, First Dose, Second Dose, Male, Female) are present	Pass
TC03	Handle missing values	Dataset with null values	Missing values replaced or handled correctly	Pass
TC04	First dose calculation	State-wise data	Correct total of first dose per state	Pass
TC05	Second dose calculation	State-wise data	Correct total of second dose per state	Pass
TC06	Male vaccination count	Male column	Correct total male vaccinated	Pass
TC07	Female vaccination count	Female column	Correct total female vaccinated	Pass
TC08	Data grouping	State column	Proper grouping of data by state	Pass
TC09	Visualization (Bar chart)	First dose data	Bar chart displayed correctly	Pass
TC10	Visualization (Pie chart)	Male & Female data	Pie chart displayed correctly	Pass
TC11	Invalid file path	Wrong CSV file path	Error message displayed	Pass
TC12	Missing column	Dataset without required column	Error handled gracefully	Pass

Chapter 7

Conclusion

The COVID-19 Vaccination Data Analysis project successfully demonstrates the application of data analysis techniques using Python. The system efficiently processes a real-world dataset to extract meaningful insights related to vaccination progress across different states in India.

Through this project, state-wise analysis of first dose and second dose vaccinations was performed, along with gender-wise analysis of vaccinated individuals. The use of data preprocessing techniques ensured accurate and reliable results. Visualization methods such as bar charts and pie charts enhanced the understanding of data patterns and trends.

The implementation using Python libraries like Pandas and Matplotlib proved to be effective for handling large datasets and performing computations efficiently. The project highlights the importance of data analysis in real-world scenarios, especially in the healthcare domain for monitoring and decision-making.

Overall, the system is simple, efficient, and scalable. It can be further enhanced by integrating advanced analytics, real-time data updates, and interactive dashboards. This project provides a strong foundation for applying data science techniques to solve practical problems.